

ECE 180J: Adv Renewable Energy Sources

Lecture 1a: Syllabus Overview

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Course Introduction

- Renewable energy, storage, and smart grids
- Covers:
 - ▶ Solar, wind, hydropower, geothermal, etc
 - ▶ Energy storage and electric vehicles
 - ▶ Microgrids and smart grids
- Includes MATLAB-based problem solving

Why This Course Matters

- Clean energy is critical for future society
- Applications across:
 - ▶ Engineering, environment, economics
- Builds skills for:
 - ▶ Industry careers
 - ▶ Graduate research

Learning Objectives

- Understand:
 - ▶ Power systems and electricity markets
 - ▶ Optimization and control of the grid
- Analyze:
 - ▶ Renewable generation and energy storage
 - ▶ Electric vehicles and grid integration
- Evaluate:
 - ▶ Social and economic impacts of clean energy

Course Logistics

- Time: Tues and Thurs, 9:50–11:25am
- Location: Porter Acad 148
- Platforms:
 - ▶ Canvas (materials)
 - ▶ Gradescope (submissions)
- Prerequisite:
 - ▶ Calculus and probability

Teaching Team

- Instructor: Dr. Yu Zhang (zhangy@ucsc.edu)
 - ▶ Office Hours: Tuesday 3–4pm (E2-547B)
- TA: Osman Saleem (ossaleem@ucsc.edu)
 - ▶ Office Hours: Wednesday 1–2pm (E2-244)
- Group Tutor: Jayant Dharwadkar (jdharwad@ucsc.edu)
 - ▶ Tutoring Hours: TBD

Workload

- Homework: 15%
- In-class midterm Exam (Tuesday May 19th): 45%
- Project:
 - ▶ Presentation: 20%
 - ▶ Report: 20%
- Curving:
 - ▶ Applied to the entire class, not applied individually
 - ▶ Applied at the end of the quarter, at the instructor's discretion

Letter Grading

Grade	Score Range
A+	95–100
A	90–94
A-	85–89
B+	80–84
B	75–79
B-	70–74
C	65–69
D	60–64
F	< 60

Pass/No Pass: Pass is awarded for grade C or higher.

Academic Integrity

- Submit your own solution with supporting work
- No late submissions (except approved cases)
- Grade disputes within 1 week
- No AI tools (ChatGPT, Gemini, Claude, Copilot, etc.)
 - ▶ Not allowed for homework or exams

Course Project

- Each team: 3–4 students
- Submit 1-page proposal by Tues of Week 5
- Topic:
 - ▶ Any topic in power systems
 - ▶ Sample topics provided (not restrictive)
- Resources:
 - ▶ Report template and rubric on Dropbox

Course Structure

- Module 1: Renewable Energy Sources
- Module 2: Energy Storage Techniques
- Module 3: Electric Vehicles and Microgrids
- Module 4: Project Presentations

Topics Overview

- Solar energy (thermal and photovoltaic)
- Wind energy
- Hydropower, tidal
- Biomass, geothermal
- Energy storage systems
- Electric vehicles
- Microgrids

Support

- DRC accommodations available
- Contact DRC for eligibility
- Accommodation letters via AIM system

Welcome

- Energy is one of the most impactful fields today
- This course connects theory with real-world systems
- Ask questions and stay engaged
- Looking forward to a great quarter